

HYDERABAD BEGUMPET, India

WMO: 431280

Lat: 17.452N

Lon: 78.461E

Elev: 1742

StdP: 13.79

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	56.1	58.7	35.6	32.6	85.8	39.2	37.6	85.3	11.7	79.9	10.2	79.7	0.2	0	0.331

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	20.9	105.9	71.5	103.8	71.5	101.7	71.5	79.5	89.3	78.3	88.1	77.3	87.1	6.5	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.2	151.1	83.9	75.7	143.7	82.2	75.1	140.7	81.5	44.7	89.9	43.5	88.5	42.4	87.0	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.1	14.0	12.5	50.7	108.9	2.9	2.7	48.7	110.8	47.0	112.4	45.4	113.9	43.3	115.9		
(5)			46.8	81.9	2.9	2.0	44.7	83.4	43.0	84.6	41.4	85.7	39.3	87.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	80.5	72.8	77.7	83.9	89.1	91.8	85.3	80.6	79.4	79.9	78.6	74.9	72.2
	DBStd	6.76	3.45	3.14	3.61	3.71	3.94	4.54	3.15	2.52	2.40	2.74	3.01	3.32
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	1	1	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11141	705	776	1052	1174	1296	1060	950	910	896	886	748	688
	CDD65	5667	241	356	587	724	831	610	485	445	446	421	298	223
	CDH74	65171	2132	3860	7785	10686	12800	7689	4630	3648	3818	3710	2482	1931
	CDH80	32308	650	1780	4424	6669	8450	3982	1615	992	1202	1327	718	499
Wind	WSAvg	5.3	3.7	4.3	4.3	4.4	6.1	8.5	8.6	7.6	5.5	4.1	3.6	3.2
Precipitation	PrecAvg	32.4	0.2	0.3	0.7	0.9	1.4	4.3	6.6	7.2	6.5	3.8	0.7	0.2
	PrecMax	54.5	2.0	3.2	6.5	3.9	9.2	8.9	16.6	18.5	19.1	12.8	2.3	3.7
	PrecMin	16.3	0.0	0.0	0.0	0.0	0.0	1.0	1.3	0.8	0.6	0.0	0.0	0.0
	PrecStd	7.8	0.4	0.7	1.1	0.9	1.7	2.0	3.5	4.0	3.9	3.0	0.7	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.5	97.2	104.2	108.0	109.5	105.7	94.7	91.4	92.8	92.9	89.6	91.5
		MCWB	67.1	65.7	69.2	70.2	72.0	72.8	72.7	73.6	75.4	71.0	67.5	67.1
	2%	DB	88.1	94.6	100.7	105.4	106.4	100.6	91.5	88.5	89.8	90.0	87.6	86.8
		MCWB	65.5	65.7	69.1	71.0	71.9	74.0	74.1	74.3	74.8	71.5	68.0	66.3
	5%	DB	86.0	91.7	98.6	102.6	104.3	97.2	89.6	87.0	88.0	88.2	85.9	84.4
		MCWB	64.7	66.0	68.3	71.3	71.8	74.0	74.3	74.3	74.8	70.9	68.0	65.0
	10%	DB	83.9	89.6	96.1	100.4	102.3	94.6	87.6	85.2	86.2	86.4	84.2	82.5
		MCWB	64.5	65.8	67.7	71.1	71.8	73.7	73.9	74.2	74.6	71.1	67.6	64.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.3	73.8	76.7	79.8	80.7	80.2	79.2	79.1	80.4	80.4	76.3	74.4
		MCDB	82.8	84.5	88.8	96.0	96.6	93.1	86.9	85.3	86.0	86.0	83.0	83.8
	2%	WB	70.2	71.8	74.9	77.4	78.4	78.8	77.5	77.5	78.6	77.8	73.8	72.0
		MCDB	79.6	82.3	87.7	92.6	93.9	90.5	84.9	83.6	84.7	83.6	81.8	80.3
	5%	WB	68.8	70.5	73.5	76.1	77.1	77.6	76.5	76.5	77.0	76.1	72.6	70.2
		MCDB	78.2	81.6	86.2	91.1	93.1	88.5	83.8	82.4	83.6	82.2	79.9	78.0
	10%	WB	67.6	69.2	72.1	74.9	76.0	76.5	75.8	75.7	76.0	74.7	71.5	68.8
		MCDB	77.5	81.0	85.8	89.8	92.3	87.0	83.2	81.6	82.5	80.8	78.2	76.9
Mean Daily Temperature Range	5% DB	MDBR	23.0	24.0	24.5	22.6	20.9	16.0	12.3	11.4	13.0	16.2	19.7	22.9
		MCDBR	26.0	27.8	27.5	25.2	22.8	19.8	15.3	14.1	15.7	19.1	21.8	25.7
		MCWBR	8.9	8.6	7.9	8.3	8.1	6.5	6.0	6.2	6.1	7.6	8.5	10.1
	5% WB	MCDBR	20.6	21.0	22.6	22.4	20.6	16.1	12.7	11.9	13.4	14.6	15.3	18.6
		MCWBR	8.4	8.0	8.3	8.6	8.0	6.9	6.4	6.2	6.7	7.9	7.1	9.0
Clear-Sky Solar Irradiance	taub	0.451	0.463	0.531	0.571	0.605	0.583	0.540	0.522	0.515	0.531	0.478	0.455	
	taud	2.077	2.062	1.907	1.858	1.785	1.877	2.006	2.067	2.081	1.993	2.077	2.080	
	Ebn at Noon	262	266	252	242	231	234	245	251	253	244	252	256	
	Edh at Noon	49	53	63	67	71	64	56	54	52	56	49	48	
All-Sky Solar Radiation	RadAvg	1542	1807	1967	2099	2082	1657	1397	1400	1525	1589	1531	1469	
	RadStd	71	69	102	76	73	152	119	84	103	82	110	80	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.45	N/A	-3.01	+1.73	N/A	N/A	N/A	+252	+250
(47) Regional (3 neighbors)		+0.41	N/A	N/A	+1.31	N/A	N/A	N/A	+252	+250

Nomenclature: See separate page